

Student Name: Sean Bush

Project Name: Is Poverty a Predictor of Low Literacy Rates?

Project Topic: Is poverty rate a statistically significant predictor of literacy rate in the United States on a per-county basis?

Research Question: Is there a statistically significant linear association between the poverty rate (the proportion of adult residents living below the poverty line) in a given U.S. county and the literacy score of that county, as estimated by the Program for the International Assessment of Adult Competencies (PIAAC)?

Hypothesis: Null Hypothesis- Poverty rate is not a statistically significant predictor of literacy rate.

Alternate Hypothesis- Poverty rate is a statistically significant predictor of literacy rate.

Context: In the wake of the COVID-19 pandemic and its effects on educational outcomes, both educators and lawmakers are looking for ways to improve literacy rates to meet and exceed pre-pandemic levels. These interested parties need verifiable, data-backed solutions to the so-called literacy crisis. If a significant link can be identified between poverty rate and literacy rate, it would provide significant evidence that literacy interventions should be directed at low-income communities and that policy that reduces poverty would also improve literacy greatly. An analysis using linear regression would either establish or deny this link, providing lawmakers and educators with appropriate, data-backed guidance.

Data: To address the research question, I will need to collect data on literacy rates in the U.S. on a per-county basis. Literacy scores would have to be collected by a relevant organization using a test or other method. Additionally, I will need to collect data on poverty rates in the U.S. on a per-county basis, which should be easily accessible from open government data. Lastly, I should also include other possible predictors that might be more significant predictors of literacy rate than poverty rate.

I will use an existing dataset compiled by the National Center for Education Statistics for the Programme for the International Assessment of Adult Competencies (<https://nces.ed.gov/surveys/piaac/skillsmap/>). This dataset contains literacy and numeracy scores for U.S. counties, states, and for the entire country. It also contains other demographic and geographic information for each county including population, proportion of residents below the poverty rate, proportion of residents significantly below the poverty rate, employment rate, and more.

This data is owned by the National Center for Education Statistics, an agency of the U.S. Department of Education. This website makes clear that use of the data for analysis is permitted. The Data User's Guide at the skill map website (<https://nces.ed.gov/surveys/piaac/skillsmap/>) gives a few stipulations about how the dataset should be used including not neglecting error terms and avoiding certain combinations of variables that would introduce multilinear. The Data User Guide explicitly states that the data set can support research analysis, regression, and reporting.

Data Gathering: Since I am using an existing dataset, data-gathering is straightforward. The Skills Map website provides an option to download the dataset as an excel spreadsheet file. From that downloaded dataset, I will select the relevant variables to create the dataset that will be used for the regression analysis.

Data Analytics Tools and Techniques: I will be using Linear Regression to analyze this data. Specifically, I will use the Python package StatsModels to perform the regression ordinary least squares. I will also use the package to perform a t-test on the relationship between the poverty rate predictor and the literacy score outcome variable.

Justification of Tools/Techniques: Linear regression is appropriate for this analysis since the coefficients it generates provide clear quantified evidence for the strength of the linear association between each predictor and the outcome variable. The StatsModels package is appropriate because it provides all the necessary statistical functionality for performing linear regression and performing the t-test, which will yield the p-value necessary to reject the null hypothesis and answer the research question.

Project Outcomes: There will be three main deliverables for this project. The first is a Jupyter Notebook which will provide all of the code required to create the linear regression model, perform the t-test, and identify the p-value in a completely reproducible way. The second deliverable will be a report which will present the research question, details the process used to collect the data, prepare the data, create the model, perform the t-test,

and answer the research question. The final deliverable will be a Pantopto presentation which will summarize the project's findings.

Projected Project End Date: April 17, 2026

Sources:

National Center for Education Statistics. (April, 2020). *PIAAC Skills Map*. U.S. Department of Education.
<https://nces.ed.gov/surveys/piaac/skillsmap/>

Krenzke, T., Mohadjer, L., Li, J., Erciulescu, A., Fay, R., Ren, W., Van de Kerckhove, W., Li, L., and Rao, J.N.K. (2020). Program for the International Assessment of Adult Competencies (PIAAC): *State and County Estimation Methodology Report (NCES 2020-225)*. U.S. Department of Education, National Center for Education Statistics. Washington, DC: U.S. Government Printing Office.
<https://nces.ed.gov/pubs2020/2020225.pdf>